Classifying linear transformations

Def^{n}

A **k-pair** is a pair (V, T) where:

- V is a k-vector space
- $T : V \longrightarrow V$ is a k-linear transformation

- The k-pair (V, T) is **finite dimensional** if V is finite dimensional

- An **isomorphism** of k-pairs

$$(V_1, T_1) \xrightarrow{\cong} (V_2, T_2)$$

is a k-linear isomorphism

$$\varphi : V_1 \longrightarrow V_2$$

s.t.

$$\begin{tikzcd} V_1 \arrow[r, "\varphi"] \arrow[d, "T_1"'] & V_2 \arrow[d, "T_2"] \\ V_1 \arrow[r, "\varphi"'] & V_2 \end{tikzcd}$$

$$T_2 \circ \varphi = \varphi \circ T_1$$
$$\Updownarrow$$
$$T_2 = \varphi \circ T_1 \circ \varphi^{-1}$$

Fact: (V, T) : finite dim'l k-pair

(a) Then $\exists$ an isomorphism

$$(k^n, T_A) \xrightarrow{\simeq} (V, T)$$

where $T_A : k^n \to k^n$ is a linear transformation corresponding to a matrix $A \in M_{n \times n}(k)$

(b) $A \in M_{n\times n}(k)$ is well-defined up to conjugacy by $GL_n(k)$

Pf: (a) Choose $k^n \xrightarrow[u]{\cong} V$ associated with a basis $\mathcal{B}$

Then $A = [T]_{\mathcal{B}}$ works.

$$
\begin{tikzcd}
k^n \arrow[r, "\cong", "u"'] \arrow[d, "\underbrace{u^{-1}\circ T\circ u}_{\text{rep'd by } [T]_{\mathcal{B}}}"'] & V \arrow[d, "T"] \\
k^n & V \arrow[l, "\cong"', "u^{-1}"]
\end{tikzcd}
$$

(b) Reinterpretation of Fact from Lecture 17.

Example:

$$f(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in k[x]$$

$$V = k[x]/(f(x))$$

$$T : V \xrightarrow{x} V$$

$$\mathcal{B} = \big(1 + (f(x)), \ldots, x^{d-1} + (f(x))\big)$$

$$[T]_{\mathcal{B}} = \begin{array}{c|c} \begin{array}{ccccc} 1 & x & \cdots & x^{d-1} & \end{array} & \\ \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & & \vdots & \vdots \\ \vdots & & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{d-1} \end{bmatrix} & \begin{array}{c} 1 \\ x \\ \vdots \\ \vdots \\ x^{d-1} \end{array} \end{array}$$

: 0 off diagonal

$$T\big(x^i + (f(x))\big) = \begin{cases} x^{i+1} + (f(x)), & \text{if } i < d-1 \\ x^d + (f(x)) \\ = -a_0 - a_1 x - \cdots - a_{d-1}x^{d-1} + (f(x)) \end{cases}$$

$$1 \xmapsto{T} x \mapsto x^2 \mapsto \cdots \mapsto x^{d-1} \xmapsto{T} -a_0 - a_1 x - \dots - a_{d-1}x^{d-1}$$

Def^{n} The $[T]_{\mathcal{B}}$ above will be denoted $M_{f(x)} \in M_{d\times d}(k)$

$\cdot$ $M_0 = 0$

Observation:

$$(k^d, M_{f(x)}) \xrightarrow{\simeq} \left(\frac{k[x]}{(f(x))}, \underset{\substack{\uparrow \\ \text{multiplication by } x}}{T}\right)$$

Theorem:

(1) $A \in M_{n \times n}(k)$

Then there is a unique matrix A' that is conjugate to A and is of the form

$$
\begin{array}{c c}
 & \begin{array}{cccc} \overbrace{\qquad}^{d_1} & \overbrace{\qquad}^{d_2} & & \overbrace{\qquad}^{d_r} \end{array} \\
\begin{array}{c} d_1\{ \\ d_2\{ \\ \\ d_r\{ \end{array} & \begin{bmatrix} M_{f_1(x)} & 0 & 0 & 0 \\ 0 & M_{f_2(x)} & & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & & M_{f_r(x)} \end{bmatrix} \}d_r
\end{array}
\qquad \boxed{d_1 + d_2 + \dots + d_r = n}
$$

where $f_1(x), f_2(x), \dots, f_r(x) \in k[x]$

are either polynomials with leading coeff 1

and s.t.

$$f_1(x) \mid f_2(x) \mid \dots \mid f_r(x)$$

Defn A matrix of this form is said to be in **rational canonical form**

Example:

- $f_1(x) = x$, $f_2(x) = x(x+1)$

$$= x^2 + x$$

$$= x^2 + 1 \cdot x + 0 \cdot 1$$

$$\begin{matrix} & 1 & \\ 1 & \left[\begin{matrix} M_{f_1(x)} & 0 \\ 0 & M_{f_2(x)} \end{matrix}\right] & 2 \\ & 2 & \end{matrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

• $f_1(x) = x+1, \quad f_2(x) = x(x+1)$

M_{x+1}

$$\begin{bmatrix} \boxed{-1} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

M_{x^2+x}

$$M_{x^2+x} = \begin{bmatrix} 0 & -0 \\ 1 & -1 \end{bmatrix}$$

$$\parallel$$

$$M_{x^2+\boxed{1}\cdot x+\boxed{0}\cdot 1}$$

(2) (V, T): finite dim'l k-pair

Then there exists a unique set of polynomials $f_1(x), f_2(x), \dots, f_r(x) \in k[x]$

s.t.

- $f_i(x)$ has leading coeff 1 for $i = 1, \dots, r$
- $f_1(x) \mid f_2(x) \mid \cdots \mid f_r(x)$
-

$$(V, T) \xrightarrow{\cong} \left(\frac{k[x]}{(f_1(x))} \times \frac{k[x]}{(f_2(x))} \times \cdots \times \frac{k[x]}{(f_r(x))}, T \right)$$

multiplication by x co-ordinate by co-ordinate